

SYDE 556/750

Simulating Neurobiological Systems
Lecture 6: Recurrent Dynamics

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- ▶ Content: Terry Stewart, Andreas Stöckel, Chris Eliasmith



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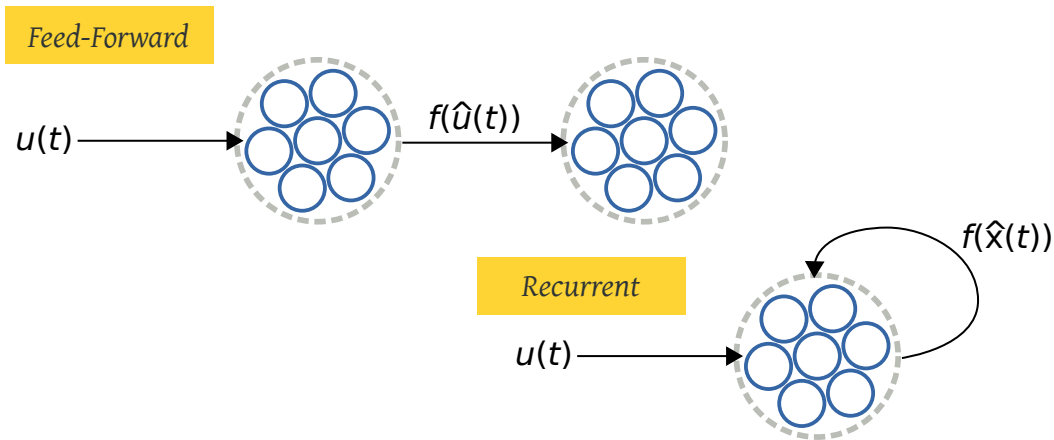


NEF Principle 3: Dynamics

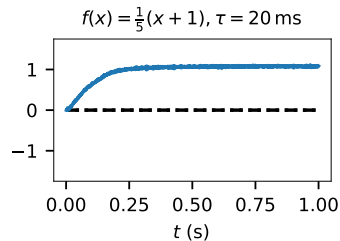
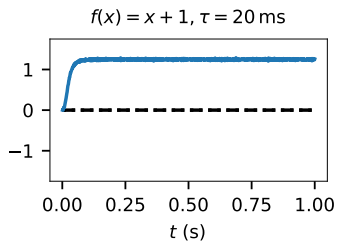
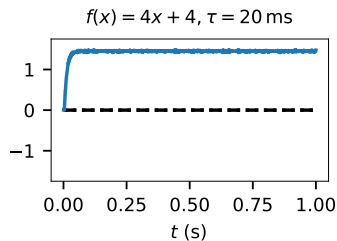
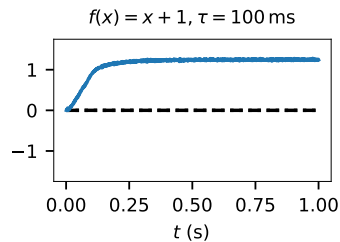
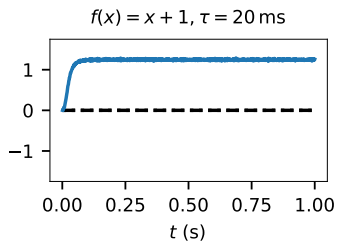
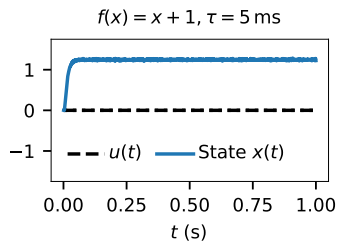
NEF Principle 3 – Dynamics

Neural dynamics are characterized by considering neural representations as control theoretic state variables. We can use control theory (and dynamical systems theory) to analyse and construct these systems.

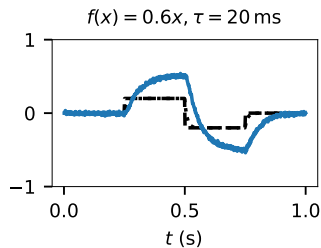
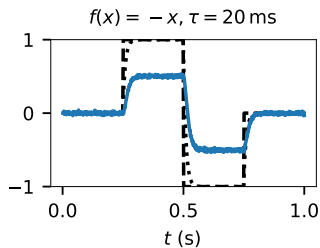
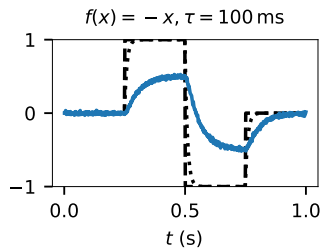
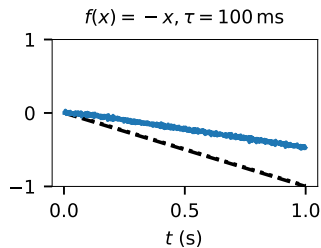
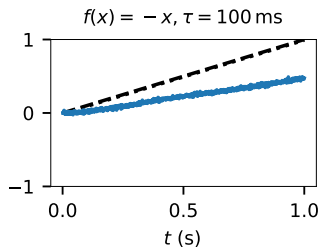
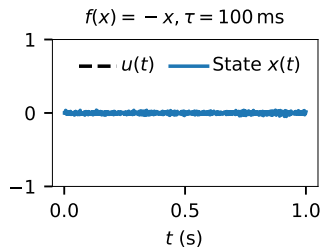
Feed Forward vs. Recurrent Connections



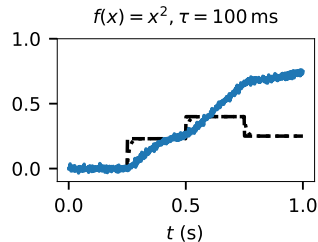
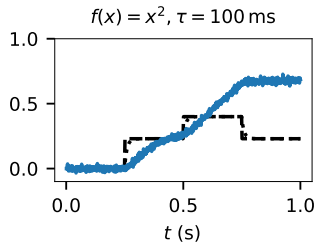
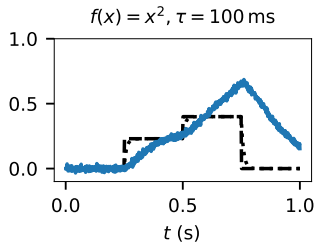
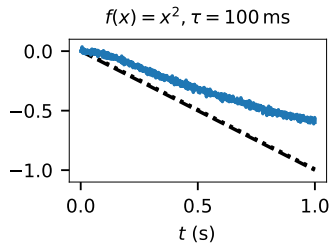
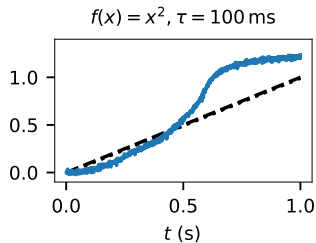
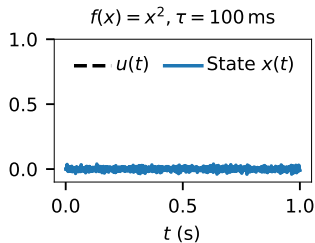
Recurrence Experiments (I)



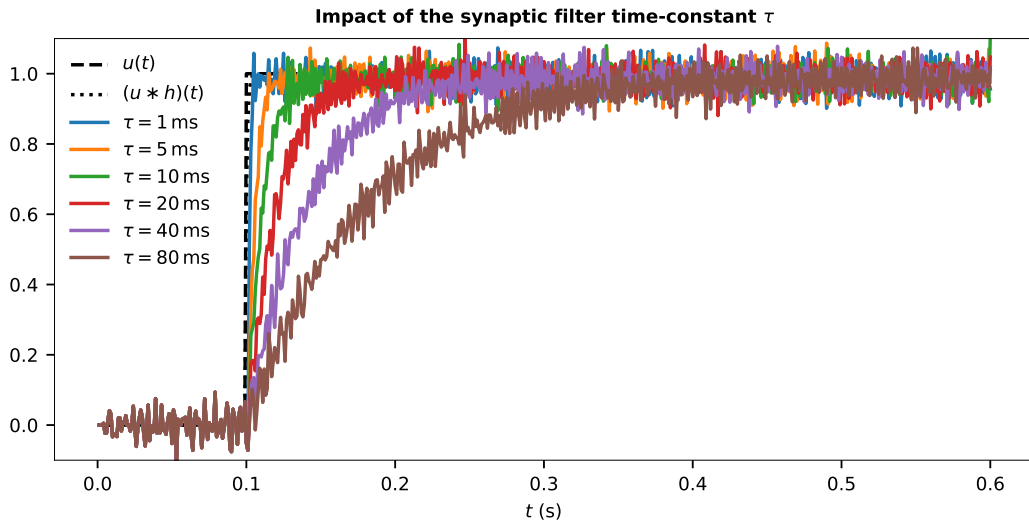
Recurrence Experiments (II)



Recurrence Experiments (III)



Making Sense of Dynamics



Behaviour of a Recurrent Connection

Feed-forward

$$x(t) = g(u(t)) * h(t)$$

Recurrent

$$x(t) = (g(u(t)) + f(x(t))) * h(t)$$

$$X(s) = (G(s) + F(s))H(s)$$

$$X(s) = (G(s) + F(s)) \frac{1}{1 + s\tau}$$

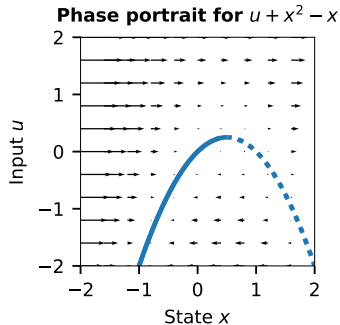
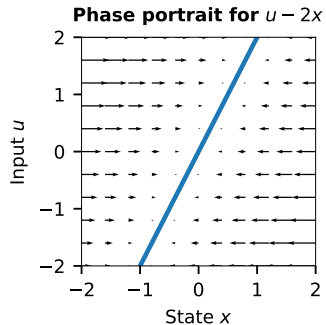
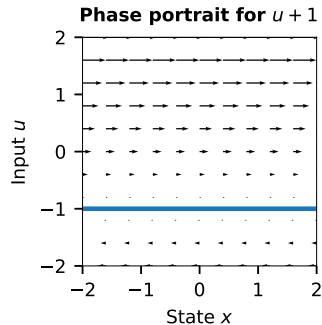
$$X(s) + s\tau X(s) = G(s) + F(s)$$

$$s\tau X(s) = G(s) + F(s) - X(s)$$

$$sX(s) = \frac{F(s) - X(s)}{\tau} + \frac{G(s)}{\tau}$$

$$\frac{dx}{dt} = \frac{f(x(t)) - x(t)}{\tau} + \frac{g(u(t))}{\tau}$$

Revisiting Recurrence Experiments



Behaviour of a Recurrent Connection

$$\frac{dx}{dt} = \frac{f(x(t)) - x(t)}{\tau} + \frac{g(u(t))}{\tau}$$

If we want this

$$\frac{dx}{dt} = a(x) + b(u)$$

Then we set

$$f(x) = \tau a(x) + x$$

$$g(u) = \tau b(u)$$

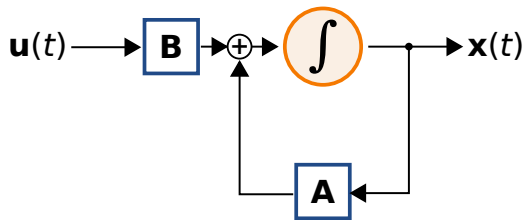
And we get

$$\frac{dx}{dt} = \frac{(\tau a(x) + x) - x}{\tau} + \frac{(\tau b(u))}{\tau}$$

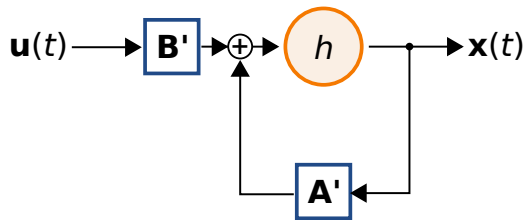
$$\frac{dx}{dt} = a(x) + b(u)$$

Implementing Dynamics using a Neural Ensemble

Evaluating an LTI
using an Integrator



Evaluating an LTI
using a Synaptic Filter



Implementing Dynamical Systems as a Neural Ensemble

LTI System

$$\varphi(\mathbf{u}, \mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\varphi'(\mathbf{u}, \mathbf{x}) = \mathbf{A}'\mathbf{x} + \mathbf{B}'\mathbf{u}$$

$$\mathbf{A}' = \tau\mathbf{A} + \mathbf{I}$$

$$\mathbf{B}' = \tau\mathbf{B}.$$

Additive Time-Invariant System

$$\varphi(\mathbf{u}, \mathbf{x}) = f(\mathbf{x}) + g(\mathbf{u})$$

$$\varphi'(\mathbf{u}, \mathbf{x}) = f'(\mathbf{x}) + g'(\mathbf{u})$$

$$f'(\mathbf{x}) = \tau f(\mathbf{x}) + \mathbf{x}$$

$$g'(\mathbf{u}) = \tau g(\mathbf{u})$$

“General” Recipe

Scale the original dynamics by τ , add feedback $\mathbf{x}$

NEF Principle 3: Dynamics

Time-Invariant Dynamical System

$$\frac{d\mathbf{x}(t)}{dt} = f(\mathbf{x}(t), \mathbf{u}(t))$$

Linear Time-Invariant (LTI) Dynamical System

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

NEF Principle 3 – Dynamics

Neural dynamics are characterized by considering neural representations as control theoretic state variables. We can use control theory (and dynamical systems theory) to analyse and construct these systems.

Low-pass Filter Example

Desired behaviour

$$\frac{dx}{dt} = \frac{u - x}{\tau_{desired}}$$

Feed-forward input

$$g(u) = \tau_{synapse} \left(\frac{u}{\tau_{desired}} \right)$$

$$g(u) = \frac{\tau_{synapse}}{\tau_{desired}} u$$

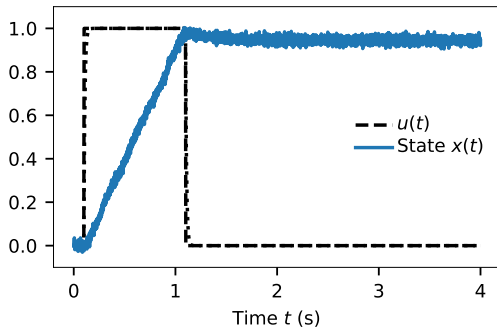
Recurrent

$$f(x) = \tau_{synapse} \left(\frac{-x}{\tau_{desired}} \right) + x$$

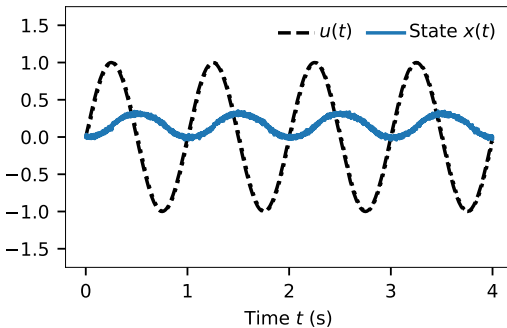
$$f(x) = \left(1 - \frac{\tau_{synapse}}{\tau_{desired}} \right) x$$

Integrator Example (I)

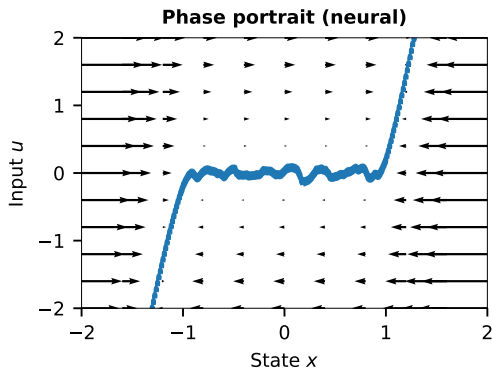
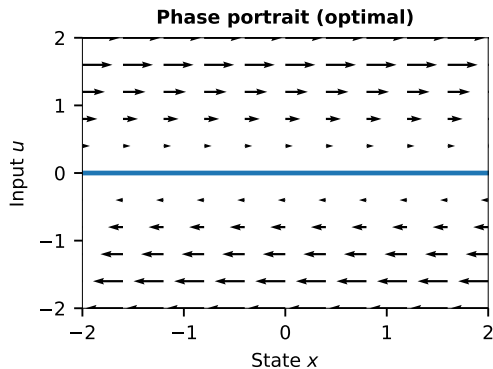
Step function input



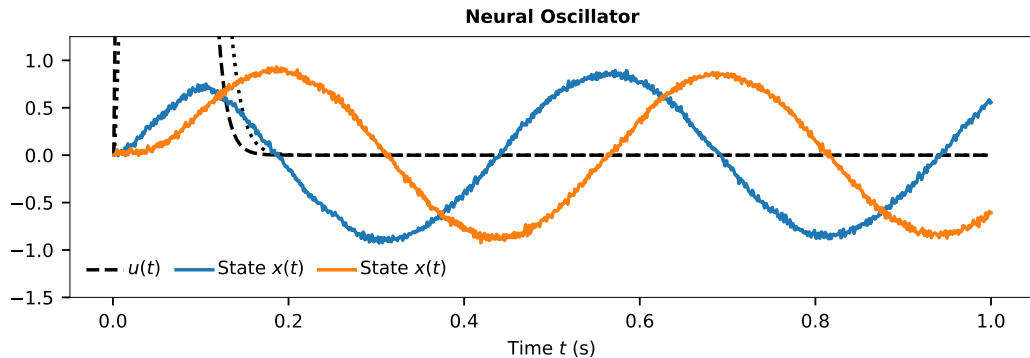
Sine input



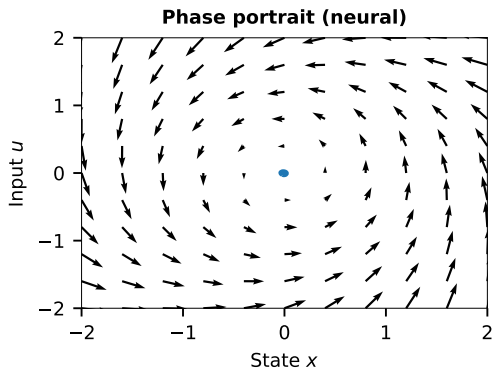
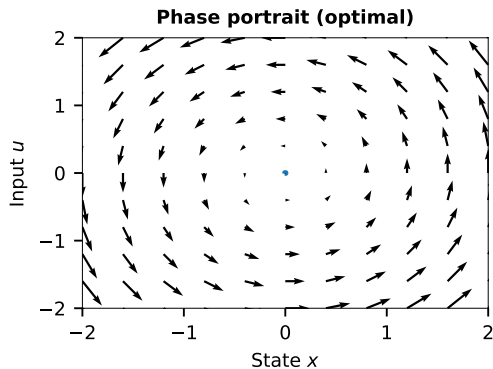
Integrator Example (II)



Oscillator Example (I)

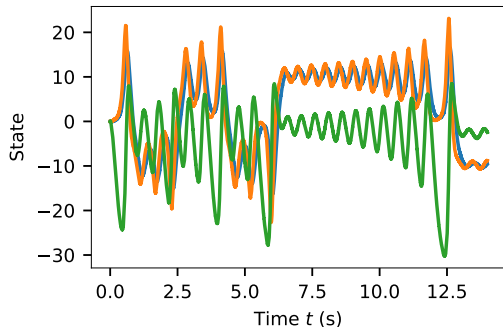


Oscillator Example (II)

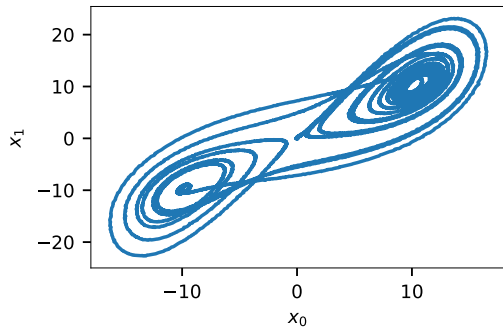


Lorentz Attractor

State over time



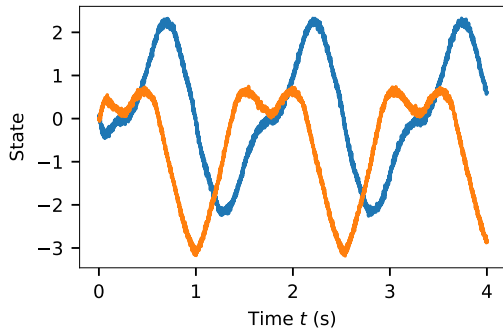
2D slice of the state space



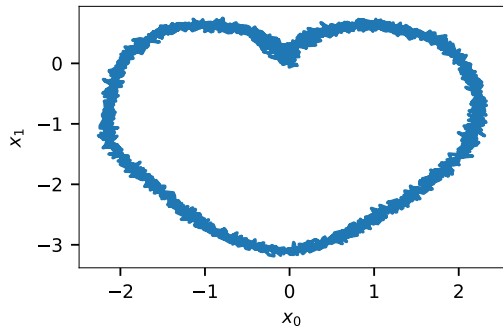
$$\frac{d\mathbf{x}(t)}{dt} = \begin{pmatrix} 10x_2(t) - 10x_1(t) \\ -x_1(t)x_3(t) - x_2(t) \\ x_1(t)x_2(t) - \frac{8}{3}(x_3(t) + 28) - 28 \end{pmatrix}$$

Heart Shape

State over time



2D slice of the state space



Horizontal Eye Control

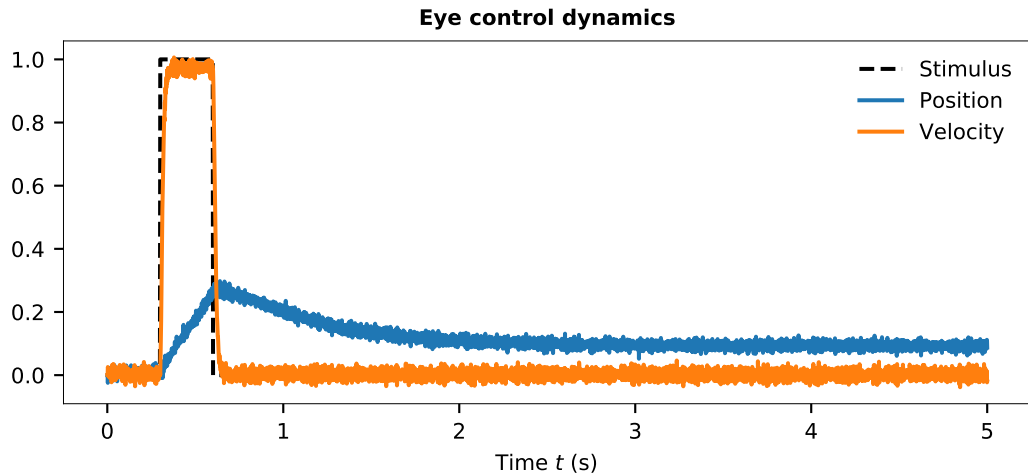


Image sources

Title slide

“The Canada 150 Mosaic Mural”

Author: Mosaic Canada Murals.

From Wikimedia.